

Quick Action Plan: Next Steps to SAO-Like World

Your Current Position

■ **Velowen.art**: Working 2D → 3DGS converter ■ **Research**: Comprehensive collection of latest papers ■ **Gap**: Static scenes, no characters, no interaction

The 5 Technology Bridges You Need

Gap	Current State	Target State	Key Technology	Readiness
1. Static → Dynamic	Frozen 3D snapshots	Animated, living worlds	4D Gaussian Splatting	8/10
2. Single → Unbounded	One scene per image	Infinite explorable worlds	SceneDreamer	8/10
3. View → Interactive	Look-only	Walk, touch, fight	Physics + SAGE-3D	7/10
4. Isolated → Persistent	Each gen independent	Coherent, evolving world	World Models	6/10
5. Silent → Social	Empty scenes	AI characters + multiplayer	GaussianAvatars + LLM	7/10

Top 10 Technologies from Your Research (Priority Order)

■ CRITICAL - Start These NOW

1. Mip-Splatting

- **Repo:** niujinshuchong/mip-splatting
- **What:** Anti-aliasing for crisp anime rendering
- **Why:** Your current 3DGS probably has shimmering artifacts
- **Time:** 1 week to integrate
- **Impact:** Professional-quality rendering

2. GaussianEditor

- **Repo:** buaacyw/GaussianEditor
- **What:** Real-time scene editing (2-7 min)
- **Why:** Users need to customize worlds
- **Time:** 2 weeks to integrate
- **Impact:** 10x user engagement

3. GaussianAvatars

- **Repo:** ShenhanQian/GaussianAvatars
- **What:** Rigged, animatable characters
- **Why:** Can't have anime world without characters
- **Time:** 1 month to get first character working
- **Impact:** Core feature for "entering anime world"

■ HIGH PRIORITY - Next Phase (Months 2-4)

4. 4D Gaussian Splatting

- **Repo:** hustvl/4DGaussians
- **What:** Add temporal dimension for animation
- **Why:** Static = boring, need motion
- **Time:** 1 month to train first animated scene
- **Impact:** Scenes feel alive

5. SceneDreamer

- **Repo:** FrozenBurning/SceneDreamer
- **What:** Generate unbounded 3D worlds from 2D images

- **Why:** Users want to explore beyond initial frame
- **Time:** 6 weeks to adapt for anime
- **Impact:** Infinite exploration

6. Gaussian Splatting SLAM

- **Paper:** Recent (2025)
- **What:** Real-time 3D reconstruction (769 FPS!)
- **Why:** Build world as users explore
- **Time:** 1 month to integrate
- **Impact:** Dynamic world expansion

■ IMPORTANT - Foundation (Months 4-6)

7. AnimeDiffusion

- **Approach:** Fine-tune Stable Diffusion on anime
- **What:** Text → anime-style 3D models
- **Why:** Let users create characters by description
- **Time:** 2 months (needs dataset + training)
- **Impact:** Democratize character creation

8. Material-Informed Gaussian Splatting

- **Paper:** Digital Twin focus (2025)
- **What:** Extract material properties for physics
- **Why:** Objects need to behave realistically
- **Time:** 1 month to prototype
- **Impact:** Real interaction (swords clashing, etc.)

9. Multimodal World Simulation

- **Paper:** Recent (2025)
- **What:** Vision + language + action integration
- **Why:** "Make it rain" should change weather
- **Time:** 2 months (complex)

- **Impact:** Natural language control

10. Nerfstudio

- **Repo:** [nerfstudio-project/nerfstudio](https://github.com/nerfstudio-project/nerfstudio)
- **What:** Modular framework for neural rendering
- **Why:** Swiss army knife for 3D experimentation
- **Time:** 2 weeks to setup
- **Impact:** Rapid prototyping

Your First 30 Days (Step-by-Step)

Week 1: Quality Upgrade

Goal: Make Veloven rendering production-quality



Deliverable: Alias-free anime 3DGS rendering

Week 2: Interactive Editing

Goal: Let users edit generated scenes



Deliverable: Web demo with real-time scene editing

Week 3: Anime Training Pipeline

Goal: Build scalable anime dataset



Deliverable: Anime-optimized 3DGS model

Week 4: First Demo

Goal: End-to-end MVP demo



Deliverable: Public demo + feedback from 100+ users

Critical Dependencies

Software:

- **CUDA 11.6+** (for 3DGS)
- **PyTorch 2.0+** (for training)
- **Python 3.8-3.10** (compatibility)
- **COLMAP** (for SfM preprocessing)
- **FFmpeg** (for video processing)

Hardware:

Minimum: - GPU: RTX 3080 (10GB VRAM) - RAM: 32GB - Storage: 500GB SSD

Recommended (for production): - GPU: RTX 4090 or A100 (24GB+ VRAM) - RAM: 64GB - Storage: 2TB NVMe SSD

Cloud Options: - **Modal:** \$1.50/hr for A100 (best for inference) - **RunPod:** \$0.69/hr for RTX 4090 (best for training) - **Lambda Labs:** \$1.10/hr for A100

Key Metrics to Track

Phase 1 (Quality):

- **FPS:** >60 at 1080p
- **PSNR:** >30 dB (photometric quality)
- **User Satisfaction:** >4/5 stars

Phase 2 (Dynamics):

- **Animation FPS:** 60 (no compromise)
- **World Size:** >1 km² explorable
- **Generation Time:** <5 min

Phase 3 (Characters):

- **Character Library:** 10+ unique rigged characters
- **Generation Time:** <2 min per character
- **Animation Quality:** Natural movement (user survey)

Resource Allocation (If You Have Budget)

Solo Developer (You):

- **Focus:** Core tech (Mip-Splatting, GaussianEditor, 4DGS)
- **Outsource:** UI/UX design, dataset collection
- **Budget:** \$5K-10K for cloud GPUs + contractors

Small Team (2-3 people):

- **You:** ML/graphics (3DGS, training)
- **Engineer #2:** Full-stack (web app, API, DB)
- **Contractor:** 3D artist (anime characters, testing)
- **Budget:** \$20K-30K/month (salaries + infra)

Funded Startup (5+ people):

- **You:** CTO/vision
- **2x ML Engineers:** 3DGS, character animation

- **1x Full-stack:** Web/mobile app
- **1x Game Dev:** Unity/Unreal integration
- **1x Designer:** UI/UX
- **Budget:** \$80K-120K/month

Common Pitfalls to Avoid

Technical:

■ **Over-optimizing too early** → Build feature-complete first, optimize later ■ **Using NeRF instead of 3DGS** → 3DGS is 100x faster for real-time ■ **Training on photorealistic data** → Anime style will be lost ■ **Ignoring FPS** → VR needs 90 FPS, plan for it from day 1

Business:

■ **Building in isolation** → Share early, get feedback ■ **Perfectionism** → Ship 80% solution, iterate based on users ■ **Ignoring copyright** → Use user-generated content, not copyrighted anime ■ **Underestimating costs** → GPU bills add up fast, plan budget

Personal:

■ **Burnout** → This is a 2-3 year journey, pace yourself ■ **Scope creep** → Focus on one phase at a time ■ **Analysis paralysis** → Start coding this week, not next month

When to Pivot / Stop

Red Flags:

- **3DGS consistently < 30 FPS** → Consider hybrid approach (mesh + splats)
- **Anime style lost in 3D** → Invest more in custom training
- **No user interest after 6 months** → Pivot to different use case (VTubers?)
- **Compute costs unsustainable** → Optimize or change model

Green Lights:

- **Users asking "when can I pay?"** → Monetize ASAP
- **Other teams copying you** → You're onto something big

- **Viral demos** → Double down on marketing
 - **VC interest** → Consider fundraising
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Your Competitive Edge

1. **First-mover in anime 3DGS** (no one else doing this)
2. **Technical depth** (you understand the papers)
3. **Existing product** (Velowen is proof of execution)
4. **Passion** (you want to live in anime worlds)

The market is waiting. Start this week. ■

Emergency Contacts (If You Get Stuck)

Academic:

- **Original 3DGS authors** (INRIA): Open to collaboration
- **Nerfstudio Discord**: Helpful community
- **SIGGRAPH/CVPR authors**: Often respond to emails

Industry:

- **Modal team**: Great support for GPU inference
- **Replicate**: Alternative for hosting models
- **Unity devs**: Check forums for VR integration

Community:

- **r/MachineLearning**: Technical questions
 - **r/anime**: User feedback
 - **Twitter**: Follow @_akhaliq for latest papers
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Final Checklist (Print This Out)

Week 1: - ☐ Fork mip-splatting repo - ☐ Integrate anti-aliasing into Velowen - ☐ Test on 10 anime images - ☐ Measure FPS improvement

Week 2: - ☐ Fork GaussianEditor repo - ☐ Setup WebUI demo - ☐ Fine-tune on 100 anime edits - ☐ User test with 5 friends

Week 3: - ☐ Scrape 10K anime images (Danbooru) - ☐ Generate multi-view data - ☐ Train anime-specific 3DGS - ☐ Document learnings

Week 4: - ☐ Build end-to-end demo - ☐ Record demo video - ☐ Post to social media - ☐ Collect 100+ user feedback

Month 2: - ☐ Prototype 4DGS animation - ☐ Test unbounded world generation - ☐ Recruit co-founder / first hire - ☐ Start fundraising outreach

Your SAO world starts TODAY. Not tomorrow. TODAY. ■■

Generated from research dated 2026-01-07. Update as new papers emerge.